

# SIMATS ENGINEERING

## ASSESSMENT TOOL 3

**COURSE NAME:** Artificial Intelligence

**COURSE CODE:** CSA1709

|                        |                         |
|------------------------|-------------------------|
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### COURSE OUTCOME COVERED

**CO1:** Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

*Assessment Tool Weightage: 10%*

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**QUESTIONS:2**

**Total Marks:20**

### Descriptive Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

**Code of Conduct:**

*I **B Santhosh Kumar/192511053** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

*B Santhosh Kumar*

**Signature of the student:**

**RUBRICS FOR CALCULATION:**

| Criterion                    | Level 4<br>(Exemplary)                                                  | Level 3<br>(Proficient)                          | Level 2<br>(Developing)                               | Level 1<br>(Beginning)                            | Max<br>Marks |
|------------------------------|-------------------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------------|---------------------------------------------------|--------------|
| Criteria                     | Level 4<br>(Exemplary)                                                  | Level 3<br>(Proficient)                          | Level 2<br>(Developing)                               | Level 1<br>(Beginning)                            | Max<br>Marks |
| Understanding<br>of Concepts | Demonstrates<br>thorough,<br>accurate<br>understanding<br>of concepts   | Good<br>understanding<br>with minor gaps         | Basic<br>understanding<br>with some<br>misconceptions | Poor<br>understanding;<br>major<br>misconceptions | 5            |
| Explanation                  | Detailed, well-<br>explained<br>answers with<br>relevant<br>examples    | Clear<br>explanation<br>with adequate<br>details | Limited<br>explanation;<br>lacks depth                | Poor or unclear<br>explanation                    | 5            |
| Application                  | Effectively<br>applies<br>concepts with<br>strong analytical<br>ability | Applies<br>concepts with<br>minor errors         | Limited<br>application with<br>weak analysis          | No application<br>or incorrect<br>analysis        | 4            |
| Organization                 | Well-structured<br>answers with<br>logical flow and<br>coherence        | Organized with<br>minor issues in<br>flow        | Basic structure,<br>lacks clarity                     | Poor<br>organization,<br>incoherent<br>answers    | 3            |
| Accuracy                     | Completely<br>accurate and<br>covers all<br>required points             | Mostly accurate<br>with minor<br>omissions       | Partially correct<br>with missing<br>points           | Incorrect<br>answers with<br>major gaps           | 2            |
| Clarity                      | Clear, neat,<br>professional<br>presentation                            | Understandable<br>with minor<br>issues           | Limited clarity,<br>readability<br>issues             | Poorly written,<br>difficult to<br>understand     | 1            |

## QUESTION 1

### Intelligent Agents in a Smart Home Automation System

#### Aim:

To analyze how different types of intelligent agents operate in a smart home automation system and identify the most suitable agent for providing user comfort, energy efficiency, and security.

#### Introduction:

A smart home automation system is an intelligent system that automatically controls home appliances such as lights, air conditioners, fans, and security devices based on user activities and environmental conditions. It continuously receives information from sensors like motion detectors, temperature sensors, light sensors, and security cameras. Based on these inputs, the system makes decisions without requiring constant human intervention.

Artificial Intelligence enables the system to learn user preferences, respond to real-time situations, and improve its performance over time. Different intelligent agents use different decision-making techniques, making them suitable for different levels of automation.

#### Smart Home Environment:

The smart home consists of various sensors, intelligent controllers, and connected devices.

| Component          | Purpose                    |
|--------------------|----------------------------|
| Motion Sensor      | Detects human movement     |
| Temperature Sensor | Monitors room temperature  |
| Light Sensor       | Measures ambient light     |
| Smart Lights       | Automatic lighting control |
| Smart Thermostat   | Controls room temperature  |
| Security Camera    | Detects intruders          |
| Door Lock          | Controls home access       |

## **Types of Intelligent Agents:**

### **1. Simple Reflex Agent**

A Simple Reflex Agent makes decisions only based on the current input without considering previous events. It follows predefined condition–action rules.

#### **Working in Smart Home**

- Motion detected → Turn ON lights
- No motion → Turn OFF lights
- Temperature above limit → Switch ON air conditioner
- Door opened unexpectedly → Trigger alarm

#### **Advantages**

- Fast response
- Easy to implement
- Low computational cost

#### **Limitations**

- Cannot remember previous events
- Cannot learn user behaviour
- Performs poorly in changing environments

### **2. Model-Based Reflex Agent**

A Model-Based Reflex Agent maintains an internal model of the environment. It combines current sensor information with previously stored information before making decisions.

#### **Working in Smart Home:**

Suppose the motion sensor temporarily stops detecting movement because the user is sitting still.

Instead of turning OFF the lights immediately, the system remembers that the room was occupied a few minutes earlier and keeps the lights ON.

Similarly, if the outside temperature suddenly changes, the system adjusts indoor temperature by considering previous readings.

#### **Advantages**

- Maintains memory of previous states
- More reliable than Simple Reflex Agent
- Handles partially observable environments

#### **Limitations**

- Requires additional memory
- Slightly more complex

### **3. Goal-Based Agent**

A Goal-Based Agent selects actions that help achieve a predefined objective.

#### **Working in Smart Home:**

Possible goals include:

- Maintain room temperature at 24°C.
- Keep the home secure.
- Reduce electricity consumption.
- Ensure user comfort.

If multiple actions are available, the agent chooses the one that best satisfies the goal.

For example, if both the fan and air conditioner can reduce temperature, the agent selects the appropriate device depending on the desired outcome.

#### **Advantages**

- Goal-oriented decision making
- Flexible behaviour
- Better planning capability

#### **Limitations**

- Requires search and planning
- Higher computation time

### **4. Utility-Based Agent**

A Utility-Based Agent evaluates all possible actions using a utility function and selects the action with the highest overall benefit.

Instead of simply reaching a goal, it tries to maximize user satisfaction while minimizing cost and energy consumption.

#### **Working in Smart Home:**

Suppose:

- User wants a comfortable temperature.
- Electricity tariff is currently high.
- Windows are open.
- Outside temperature is pleasant.

Instead of switching ON the air conditioner, the agent opens smart windows and starts a ceiling fan because it provides similar comfort with lower energy consumption.

#### **Advantages**

- Produces optimal decisions

- Balances multiple objectives
- Highly suitable for intelligent automation

### **Limitations**

- Utility calculation is more complex
- Requires more processing power

### **Comparison of Intelligent Agents:**

| Agent Type    | Memory | Goal-Oriented | Learns Preferences | Suitable for Smart Home  |
|---------------|--------|---------------|--------------------|--------------------------|
| Simple Reflex | No     | No            | No                 | Basic automation         |
| Model-Based   | Yes    | No            | Limited            | Better monitoring        |
| Goal-Based    | Yes    | Yes           | Limited            | Intelligent planning     |
| Utility-Based | Yes    | Yes           | Yes                | Best overall performance |

### **Analysis of Agent Performance:**

Each intelligent agent contributes differently to the smart home system.

- The **Simple Reflex Agent** performs immediate actions based on sensor inputs but cannot adapt to changing situations.
- The **Model-Based Agent** improves reliability by remembering previous observations.
- The **Goal-Based Agent** focuses on achieving objectives such as maintaining comfort and security.
- The **Utility-Based Agent** evaluates multiple alternatives and selects the most beneficial action by considering user comfort, energy consumption, and safety simultaneously.

### **Recommended Solution:**

A **hybrid approach combining Model-Based and Utility-Based Agents** is the most effective solution for a smart home automation system.

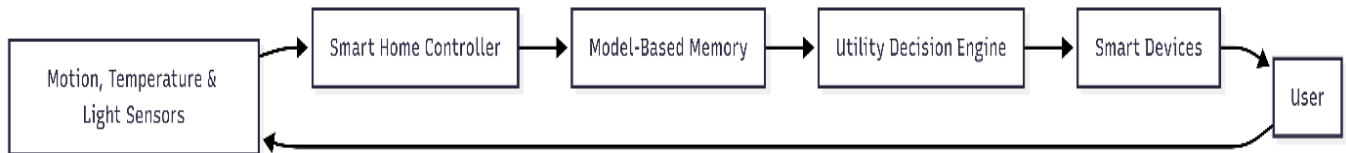
The Model-Based Agent maintains information about previous user activities and environmental conditions, while the Utility-Based Agent evaluates different alternatives and chooses the action that provides maximum comfort with minimum energy consumption.

This combination allows the system to:

- Learn user preferences over time.
- Respond to real-time sensor inputs.

- Reduce unnecessary energy usage.
- Improve home security.
- Provide better overall user experience.

**Block Diagram:**



**Figure 1: Smart Home Automation System Architecture**

**Figure Credit:** Author's illustration created using Mermaid Live Editor.

**Result:**

The smart home automation system can effectively manage lighting, temperature, and security by using intelligent agents. Among the different agent types, a **hybrid Model-Based and Utility-Based Agent** provides the best performance because it combines memory, intelligent decision-making, and optimization. This approach ensures user comfort, efficient energy utilization, and enhanced home security while adapting to changing environmental conditions.

## QUESTION 2

### Solving an Unknown Maze Using Uniform Cost Search (UCS) and Iterative Deepening Search (IDS)

#### Aim:

To analyze how Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) help a robot navigate an unknown maze, compare their performance, relate them to intelligent agents, and determine the most suitable combination for efficient path planning.

#### Introduction:

Robots operating in unknown environments must search for a safe path without having complete knowledge of the surroundings. In such situations, Artificial Intelligence search algorithms help the robot explore the environment systematically until the destination is reached. Since the robot does not know the maze layout initially, it must make decisions based on newly discovered paths while avoiding unnecessary exploration.

Among the uninformed search algorithms, **Uniform Cost Search (UCS)** and **Iterative Deepening Search (IDS)** are widely used. UCS always expands the path with the lowest accumulated cost, whereas IDS repeatedly performs depth-limited searches with increasing depth until the goal is found.

#### Problem Statement:

A robot is placed inside an unknown maze and must reach the exit without having prior knowledge of the maze layout. The robot explores the maze while avoiding dead ends and selecting suitable paths until the exit is located.

#### Assumptions:

Since the maze layout is not provided, the following assumptions are considered.

- The maze is represented as a grid.
- The robot starts at **S**.
- The exit is represented by **G**.
- Some paths are blocked.
- The robot can move only in four directions.
- Every movement has a cost of 1.

**Problem Formulation:**

| Component            | Description                                                 |
|----------------------|-------------------------------------------------------------|
| Initial State        | Robot at Start (S)                                          |
| Goal State           | Reach Exit (G)                                              |
| State Representation | Robot Position (Row, Column)                                |
| Operators            | Up, Down, Left, Right                                       |
| Search Algorithms    | Uniform Cost Search (UCS), Iterative Deepening Search (IDS) |
| Path Cost            | Cost of each movement = 1                                   |

**Working of Uniform Cost Search:**

Uniform Cost Search expands the node having the smallest cumulative path cost. Every time a node is expanded, the neighbouring nodes are inserted into the priority queue with their updated path costs. The algorithm guarantees the least-cost solution when all movement costs are positive.

**Working of Iterative Deepening Search:**

Iterative Deepening Search performs a series of depth-limited searches. Initially, it searches only one level deep. If the goal is not found, the depth limit is increased and the search is repeated until the exit is discovered.

IDS combines the low memory requirement of Depth-First Search with the completeness of Breadth-First Search.

**Comparison of UCS and IDS:**

| Feature          | Uniform Cost Search      | Iterative Deepening Search    |
|------------------|--------------------------|-------------------------------|
| Search Strategy  | Lowest path cost first   | Repeated depth-limited search |
| Complete         | Yes                      | Yes                           |
| Optimal          | Yes                      | Yes (equal costs)             |
| Time Complexity  | $O(b^{(1+C*/\epsilon)})$ | $O(b^d)$                      |
| Space Complexity | $O(b^{(1+C*/\epsilon)})$ | $O(bd)$                       |
| Memory Usage     | High                     | Low                           |
| Suitable For     | Variable path costs      | Unknown search depth          |

## **Advantages and Disadvantages:**

### **Uniform Cost Search**

#### **Advantages**

- Finds the minimum-cost path.
- Guarantees an optimal solution.
- Suitable when movement costs vary.

#### **Disadvantages**

- Requires more memory.
- Can be slower for large search spaces.

### **Iterative Deepening Search:**

#### **Advantages**

- Requires very little memory.
- Complete search method.
- Suitable for unknown search depth.

#### **Disadvantages**

- Repeats node expansion.
- Extra computation due to repeated searches.

## **Intelligent Agent Perspective:**

The robot behaves as an intelligent agent because it continuously observes the environment, selects appropriate actions, and updates its knowledge while searching for the exit.

## **Agent Components**

| <b>Component</b>    | <b>Description</b>               |
|---------------------|----------------------------------|
| Performance Measure | Reach the exit with minimum cost |
| Environment         | Unknown maze                     |
| Sensors             | Distance Sensor, Camera          |
| Actuators           | Robot Wheels                     |
| Agent Type          | Goal-Based Agent                 |

### **Effect of Agent Design on Decision Making:**

The design of the intelligent agent directly affects the robot's ability to solve the maze efficiently.

- A **Simple Reflex Agent** reacts only to immediate sensor inputs and may repeatedly enter dead ends.
- A **Model-Based Agent** remembers previously explored paths and avoids revisiting blocked locations.
- A **Goal-Based Agent** plans actions to reach the exit efficiently.
- A **Utility-Based Agent** evaluates multiple paths and selects the one that minimizes travel cost and exploration time.

### **Recommended Solution:**

A **Goal-Based Agent combined with Uniform Cost Search (UCS)** is the most suitable solution for this problem.

The Goal-Based Agent keeps the robot focused on reaching the exit, while Uniform Cost Search guarantees the minimum-cost path. If memory availability is limited, Iterative Deepening Search can also be used because it requires significantly less memory while still ensuring completeness.

### **Block Diagram:**



**Figure 1:** Intelligent Robot Navigation Using UCS and IDS

**Figure Credit:** Author's illustration created using Mermaid Live Editor.

### **Result:**

The unknown maze problem can be solved effectively using uninformed search algorithms. Uniform Cost Search guarantees the least-cost path, while Iterative Deepening Search provides a memory-efficient alternative. Among the intelligent agent types, a Goal-Based Agent integrated with Uniform Cost Search offers the best overall performance by combining systematic decision-making with optimal path selection. This approach enables the robot to reach the exit efficiently while adapting its actions based on the explored environment.